ROLL NO : 241801038

Exp 10b: Design and Configuration of an Internetwork Using Wireless Router, DHCP Server and Internet Cloud

AIM :

To design and configure an internetwork using a wireless router, a DHCP server and an Internet cloud in the Cisco Packet Tracer simulator, and to verify the connectivity between the devices.

ALGORITHM :

1. Start the experiment.
2. Launch Cisco Packet Tracer and build the topology with a PC, a Laptop, a Wireless Router (WRT300N), a Cable Modem, an Internet Cloud and the Cisco.com Server.
3. Connect the devices using copper straight-through cables (PC ↔ router, router ↔ cable modem, cloud ↔ server) and a coaxial cable (cable modem ↔ cloud).
4. Configure the wireless router: set the network name (SSID) as “HomeNetwork”, enable the DHCP server and set the DNS server as 208.67.220.220.
5. Install the wireless module (WPC300N) in the laptop and connect it to the “HomeNetwork” wireless network.
6. Set the PC to obtain its IP address automatically (DHCP) and verify it with the ipconfig /all command.
7. Configure the Internet cloud (coaxial → Ethernet port mapping and Cable provider network).
8. Configure the Cisco.com server with the DHCP service, the DNS service and a static IP address (208.67.220.220).
9. Refresh the IP settings of the PC using ipconfig /release and ipconfig /renew, and test the connectivity with ping Cisco.com.
10. Save the file and stop the experiment.

PROCEDURE :

Addressing table:

Device	Interface	IP address	Subnet mask	Default gateway
PC	Ethernet0	DHCP		192.168.0.1
Wireless Router	LAN	192.168.0.1	255.255.255.0	
Wireless Router	Internet	DHCP		
Cisco.com Server	Ethernet0	208.67.220.220	255.255.255.0	
Laptop	Wireless0	DHCP		

Part 1 – Build the network in the Logical workspace: Add the network devices by choosing the device type in the Device-Type Selection box and then the model in the Device-Specific Selection box. Change the display names of the devices (Config tab → Display Name). Add the cabling: PC FastEthernet0 → Ethernet 1 of the wireless router

(copper straight-through); Internet port of the wireless router → Port 1 of the cable modem (copper straight-through); Port 0 of the cable modem → coaxial port of the Internet cloud (coaxial); Ethernet port of the cloud → FastEthernet0 of the Cisco.com server (copper straight-through).

Part 2 – Configure the network devices:

Step 1 – Wireless router: Click the Wireless Router → GUI tab → Wireless tab and type the Network Name (SSID) as “HomeNetwork”. In the Setup tab, verify that the DHCP Server is Enabled and set the static DNS server as 208.67.220.220, then click Save Settings.

Step 2 – Laptop: Power off the laptop, remove the Ethernet copper module and install the Wireless WPC300N module, then power it on. Open Desktop → PC Wireless → Connect tab, select “HomeNetwork” and click Connect.

Step 3 – PC: Open Desktop → IP Configuration and select DHCP so that the PC receives an IPv4 address from the wireless router. In the Command Prompt, issue `ipconfig /all` to verify that the PC received an address in the 192.168.0.x range.

Step 4 – Internet cloud: Install the modules PT-CLOUD-NM-1CX (cable modem connection) and PT-CLOUD-NM-1CFE (copper Ethernet) if they are missing. In the Config tab → CONNECTIONS → Cable, choose Coaxial and Ethernet as the From and To ports and click Add. Under INTERFACE → Ethernet select Cable as the Provider Network.

Step 5 – Cisco.com server: Services tab → DHCP, with the following settings:

Setting	Value
DHCP service	On
Pool name	DHCPpool
Default gateway	208.67.220.220
DNS server	208.67.220.220
Starting IP address	208.67.220.1
Subnet mask	255.255.255.0
Maximum number of users	50

Services tab → DNS: turn the DNS service On, Name = Cisco.com, Type = A Record, Address = 208.67.220.220, and click Add. In Config → Settings select Static with Gateway 208.67.220.1 and DNS server 208.67.220.220. In Config → FastEthernet0 select Static with IP address 208.67.220.220 and subnet mask 255.255.255.0.

Part 3 – Verify connectivity: On the PC Command Prompt issue `ipconfig /release` and then `ipconfig /renew`; the PC should get an address in the 192.168.0.x range with a default gateway and DNS server. Then issue `ping Cisco.com`; four replies should be received.

OUTPUT :

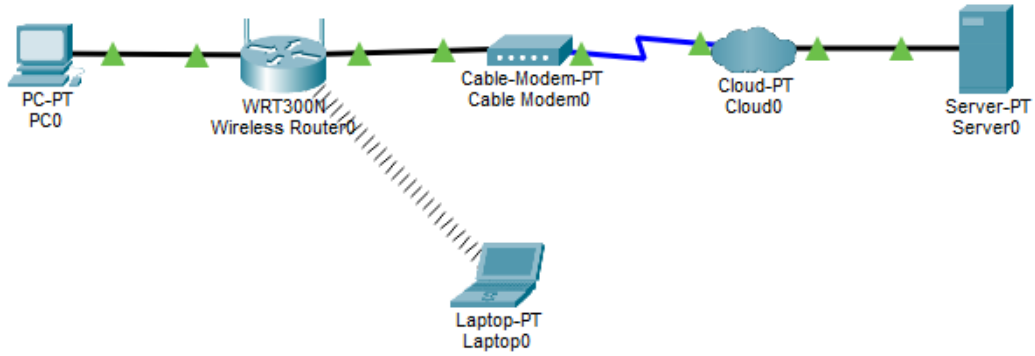


Fig 1. Designed network topology – PC, Laptop, Wireless Router, Cable Modem, Cloud and Server

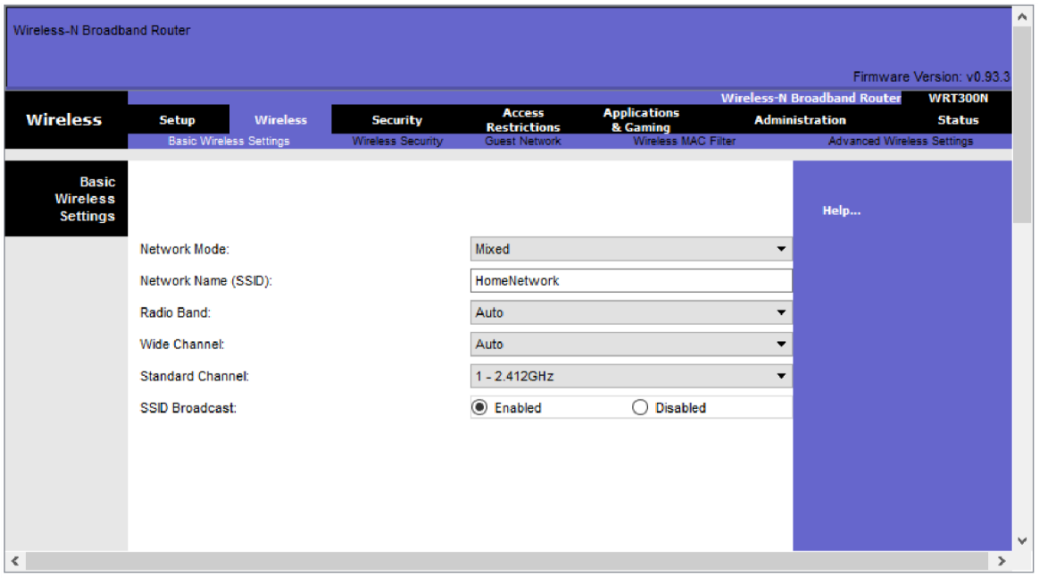


Fig 2. Wireless router GUI – Network Name (SSID) set as "HomeNetwork"

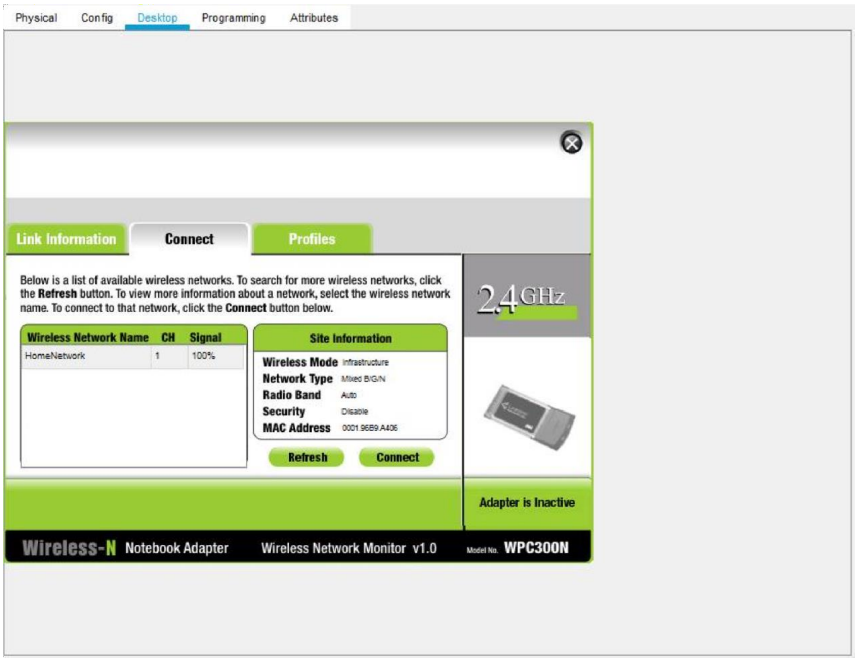


Fig 3. Laptop – connected to the "HomeNetwork" wireless network (PC Wireless → Connect)

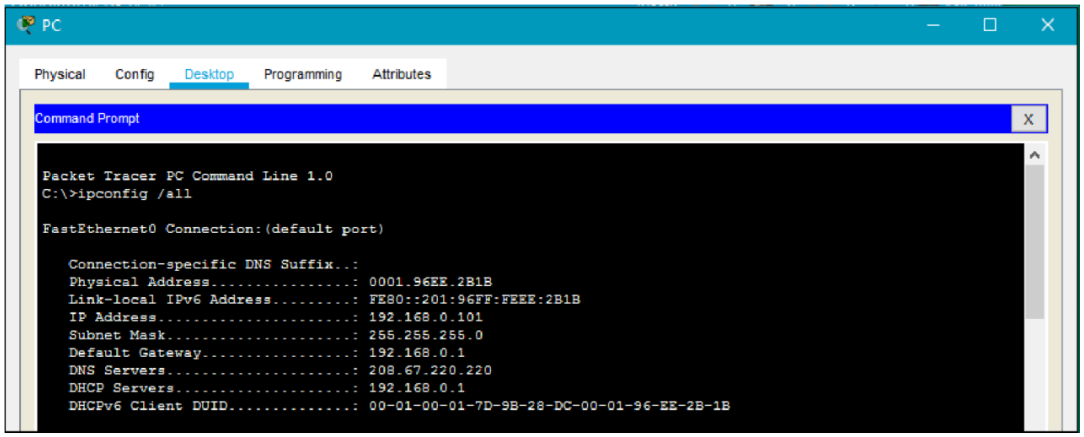


Fig 4. PC – IP address obtained through DHCP (ipconfig /all)

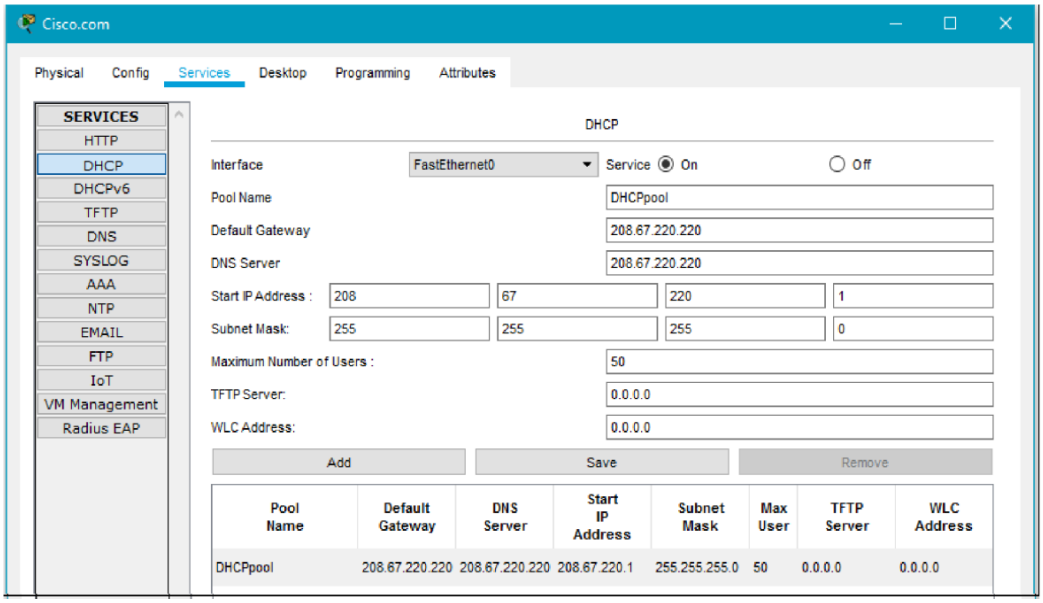


Fig 5. Cisco.com server – DHCP service and DHCP pool configuration

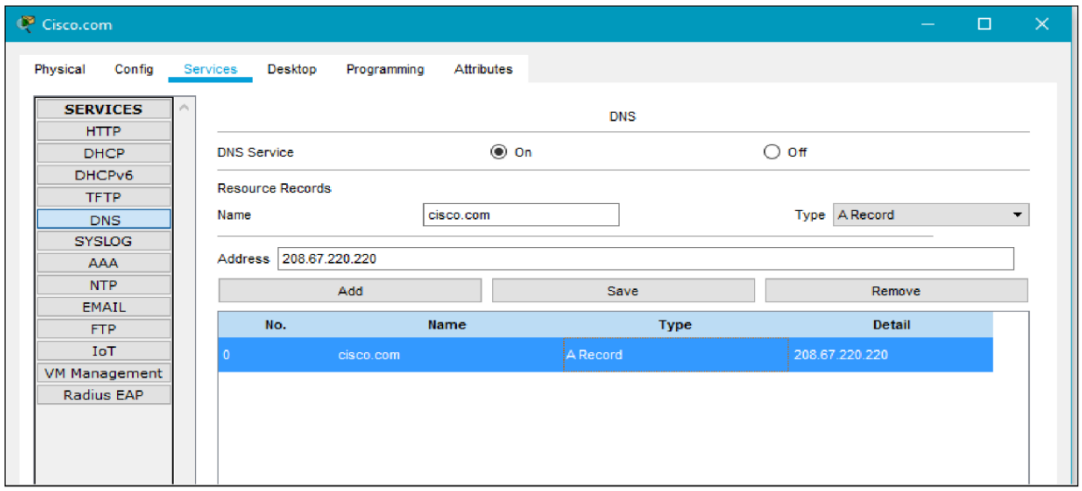


Fig 6. Cisco.com server – DNS service (cisco.com → 208.67.220.220)

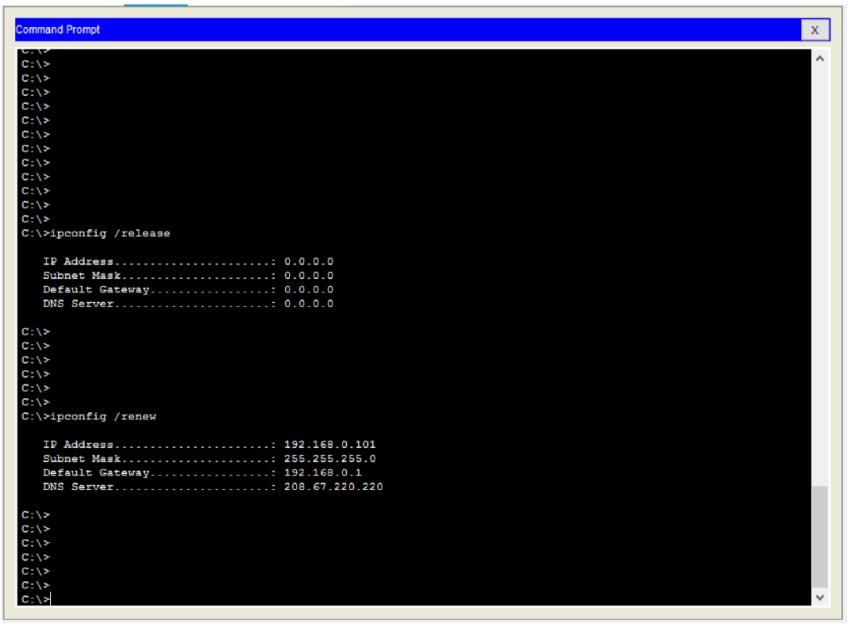


Fig 7. PC – ipconfig /release and ipconfig /renew

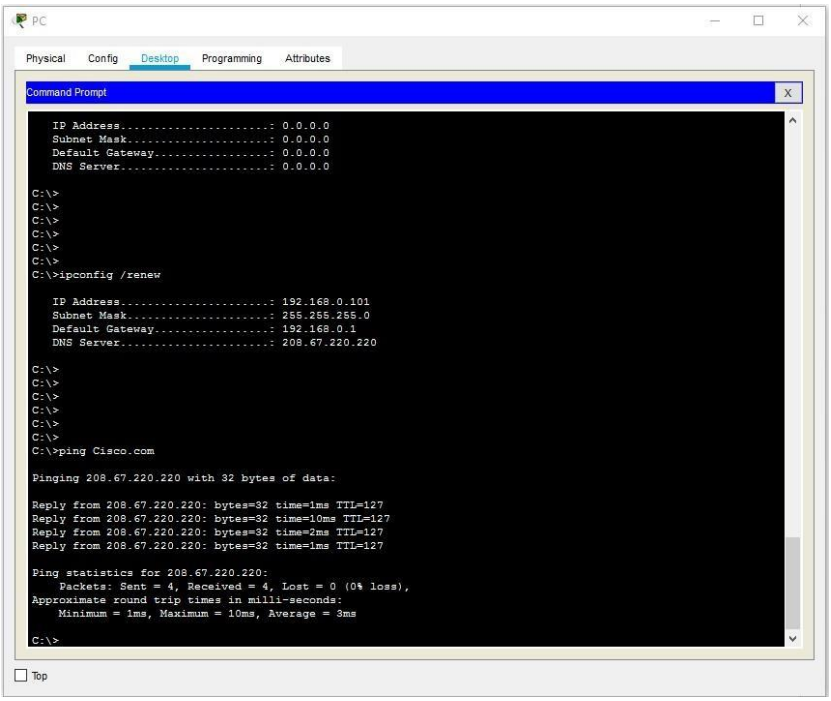


Fig 8. PC – ping Cisco.com (four replies received, 0% loss)

RESULT :

Thus, an internetwork was designed and configured using a wireless router, a DHCP server and an Internet cloud in Cisco Packet Tracer. The PC and the laptop obtained their IP addresses through DHCP, and the connectivity to the Cisco.com server was verified using the ping command.